CheXGuard AI — Thoracic Radiograph Triage Report

Scan Identifier: CASE2_PNEUMONIA | **Timestamp:** 2026-11-02 20:30:47 IST | **Model:** CheXGuardNet v2.1-DenseNet121

TRIAGE PRIORITY: ELEVATED (PRIORITY 2)

URGENCY INDEX: 0.52 / 1.00

Primary Finding: Atelectasis

Safety Validation: VERIFIED RADIOGRAPH (OOD: 0.0)

Diagnostic Predictions & Epistemic Uncertainty Bounds

Thoracic Pathology	Probability	95% Confidence Interval	Epistemic Std (σ)	Reliability Flag
Atelectasis	93.3%	[60.0%, 100.0%]	+/-0.170	[ALERT] Mandatory Specialist Verification
Cardiomegaly	80.0%	[32.0%, 100.0%]	+/-0.245	[ALERT] Mandatory Specialist Verification
Pleural Effusion	13.3%	[0.0%, 56.7%]	+/-0.221	[ALERT] Mandatory Specialist Verification
Pneumonia	13.3%	[0.0%, 56.7%]	+/-0.221	[ALERT] Mandatory Specialist Verification
Pneumothorax	13.3%	[0.0%, 56.7%]	+/-0.221	[ALERT] Mandatory Specialist Verification
No Finding	86.7%	[43.3%, 100.0%]	+/-0.221	[ALERT] Mandatory Specialist Verification

Clinical Recommendation & Next Steps

• **Recommended Protocol:** Expedited clinical correlation and priority radiologist read recommended within 2 hours.
• **Visual Grounding:** Grad-CAM class activation maps indicate localized thoracic tissue opacity corresponding to the highest probability pathology.

LEGAL & CLINICAL DISCLAIMER: *CheXGuard AI is an assistive decision-support algorithm developed for investigational triage during Neural Storm 2026. This system is not FDA/CE-cleared as a primary diagnostic device. Diagnostic decisions, patient management, and treatment plans must be made exclusively by a qualified healthcare provider.*